Detection of anomalies in time series volumetric network data

Samuel Dobron

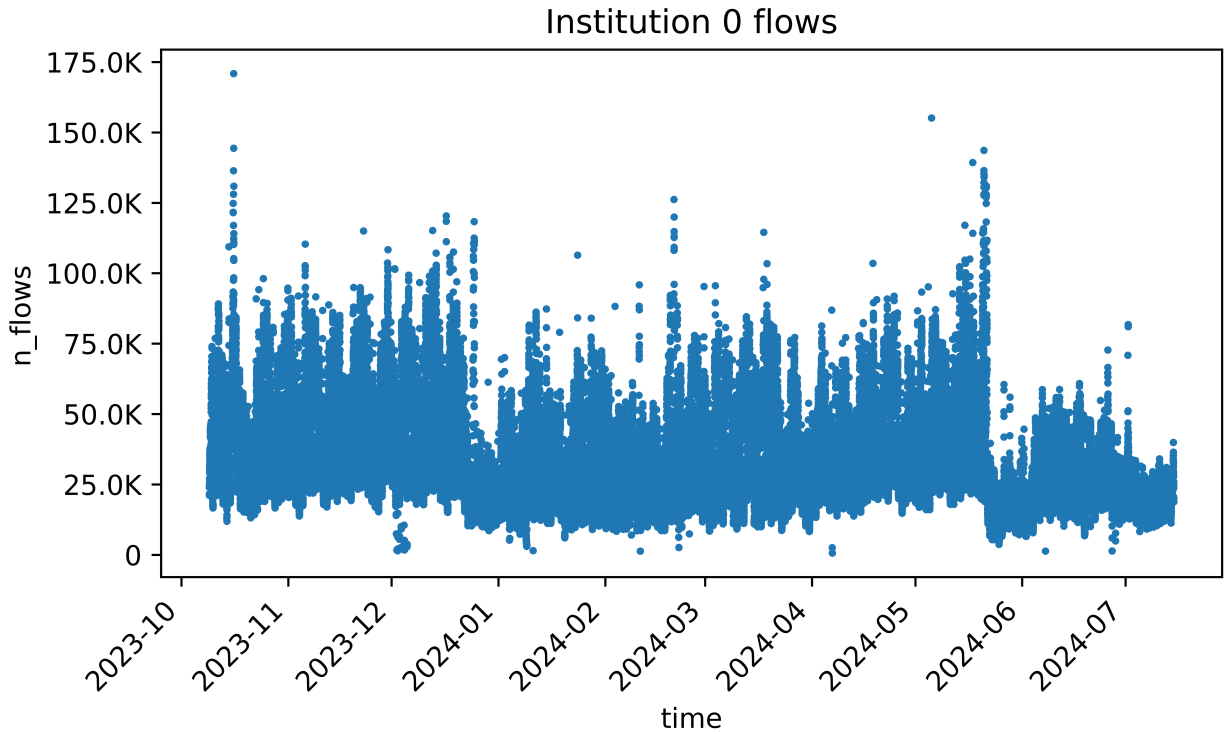
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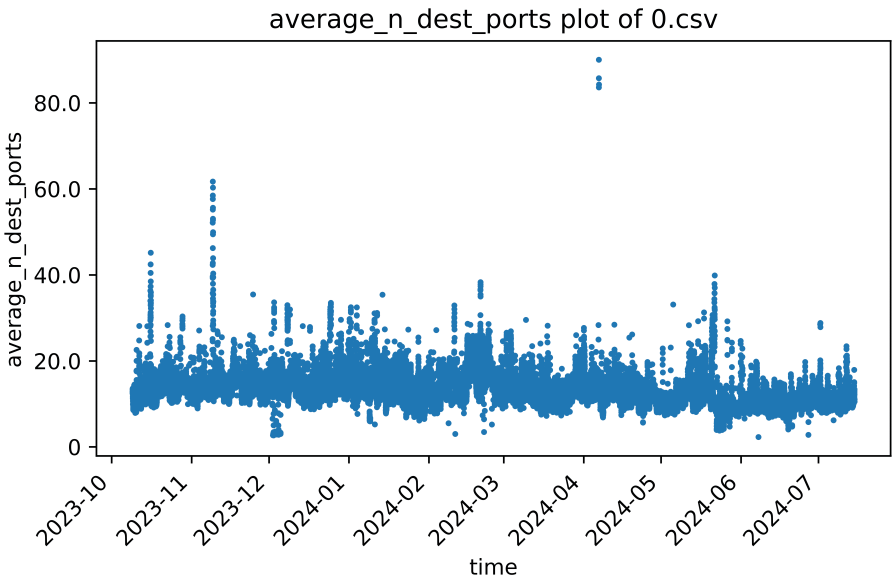
May 06, 2025

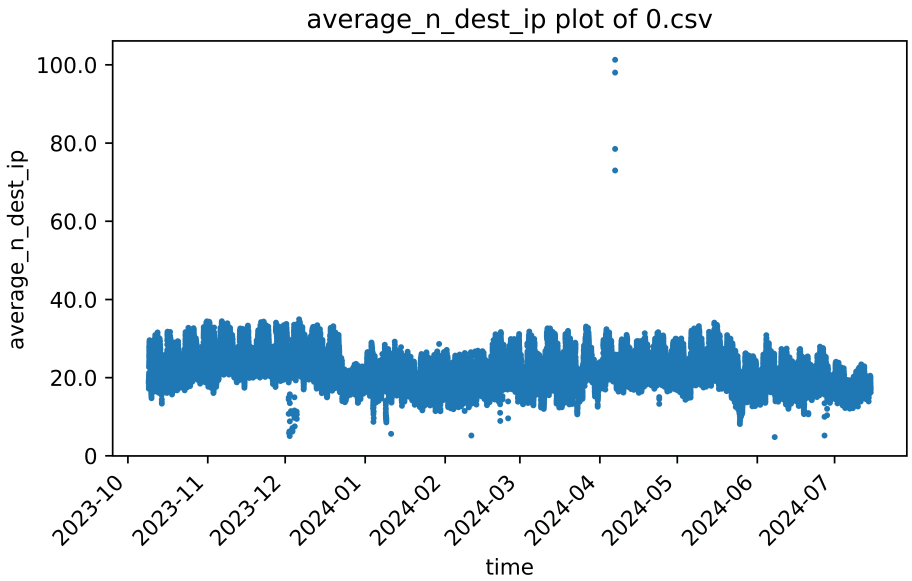
Dataset selection

- Institution 0
- Subnet 0
  - IP 1580
  - IP 230257
- Subnet 6
  - IP 11188
  - IP 299906

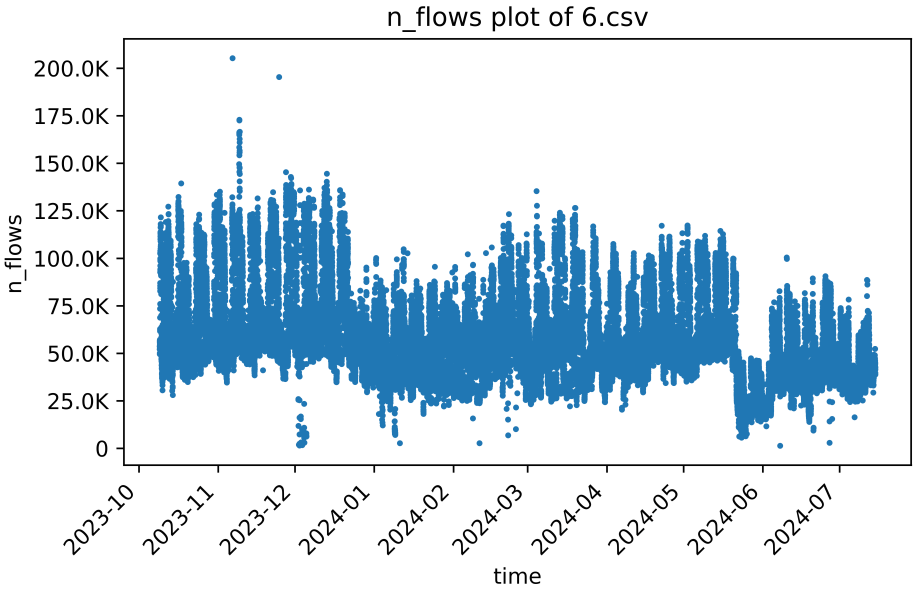


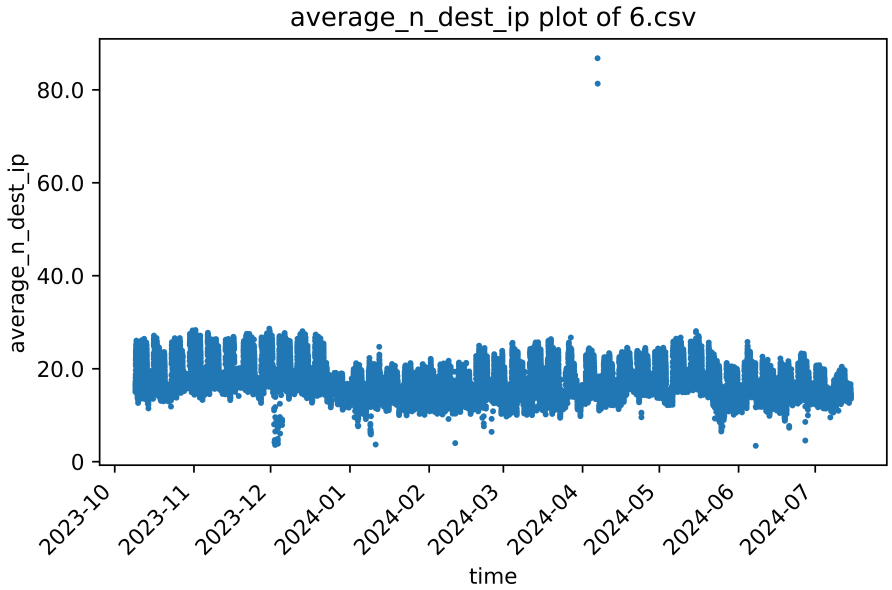
Subnet 0 anomalies

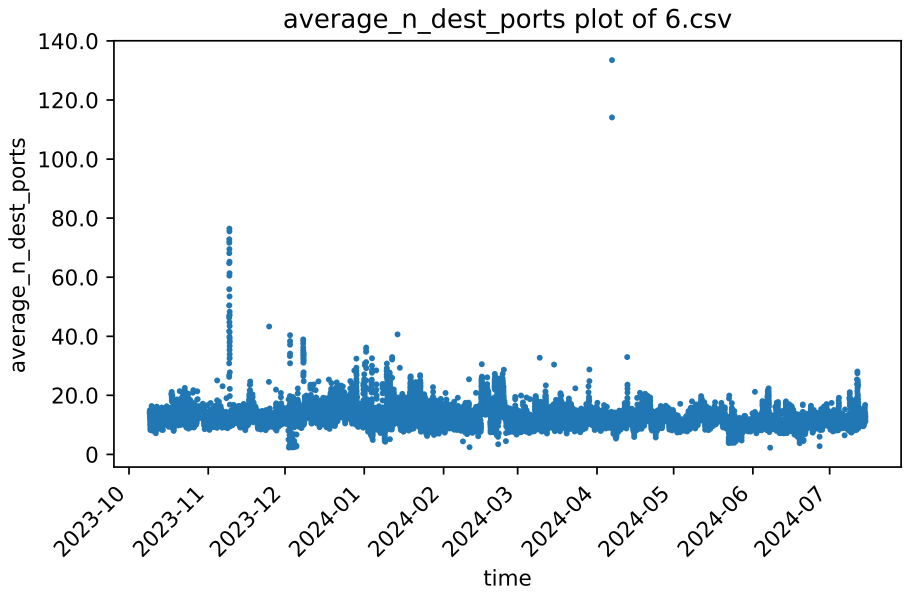


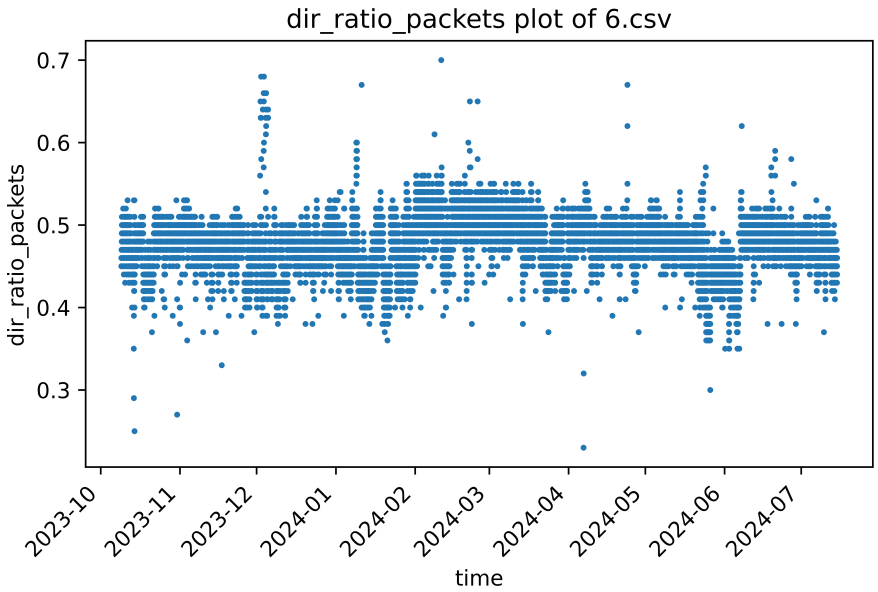


Subnet 6 anomalies

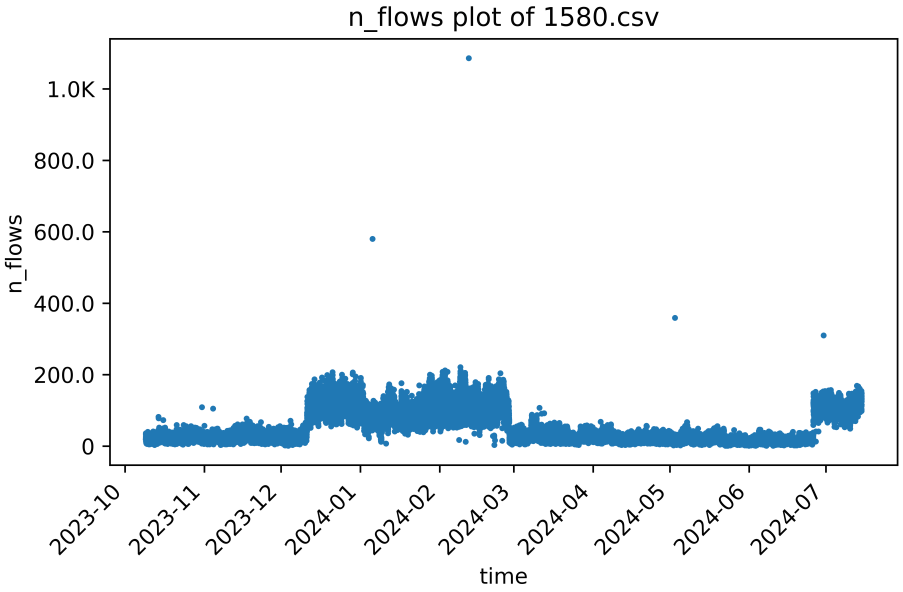


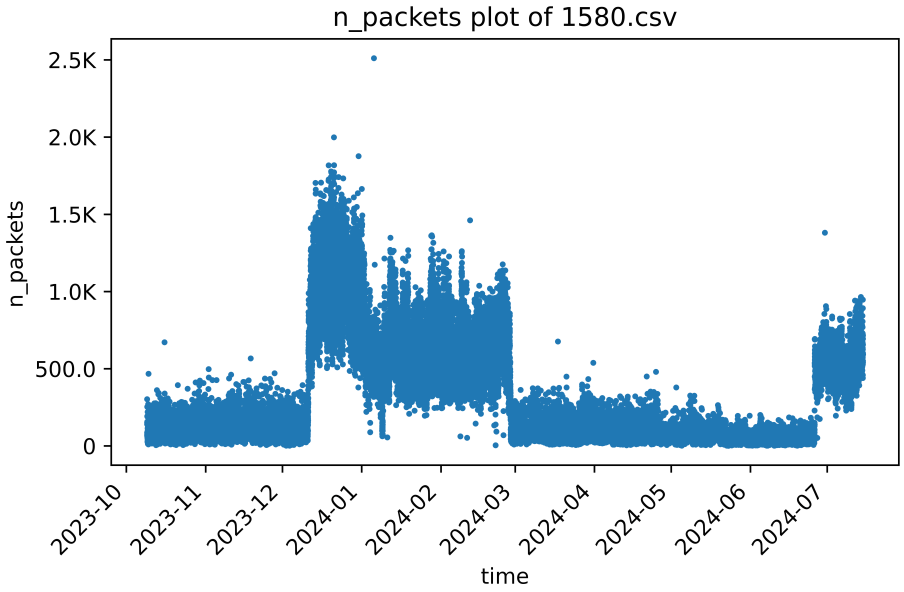


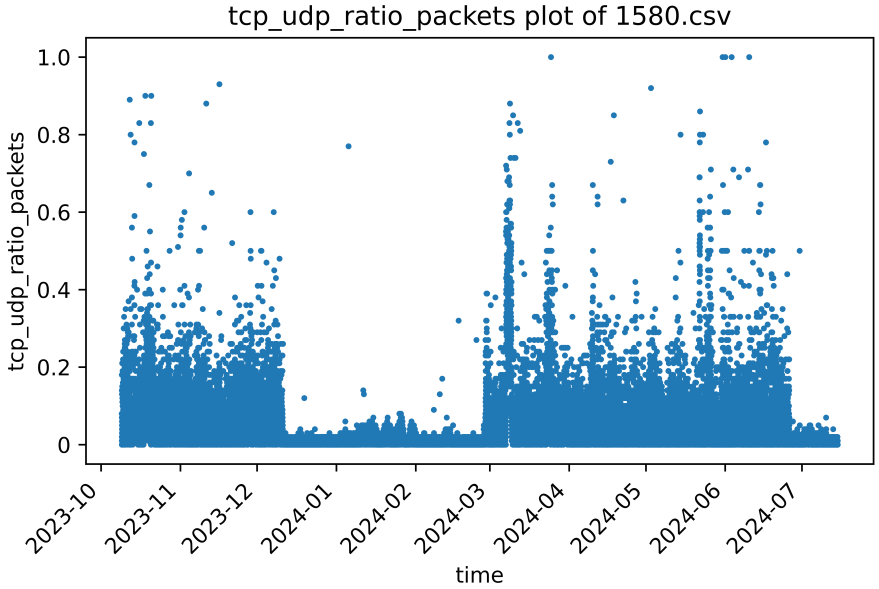


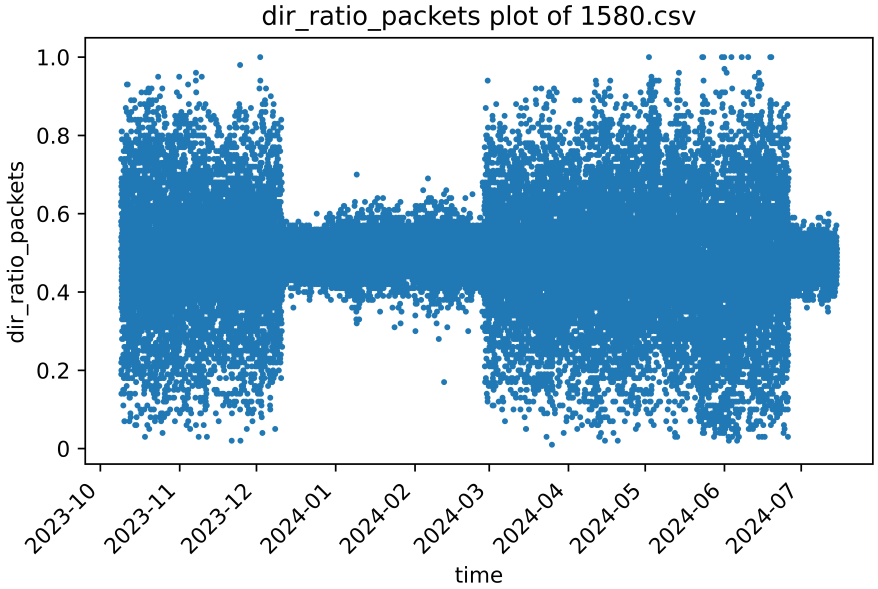


IP 1580 anomalies

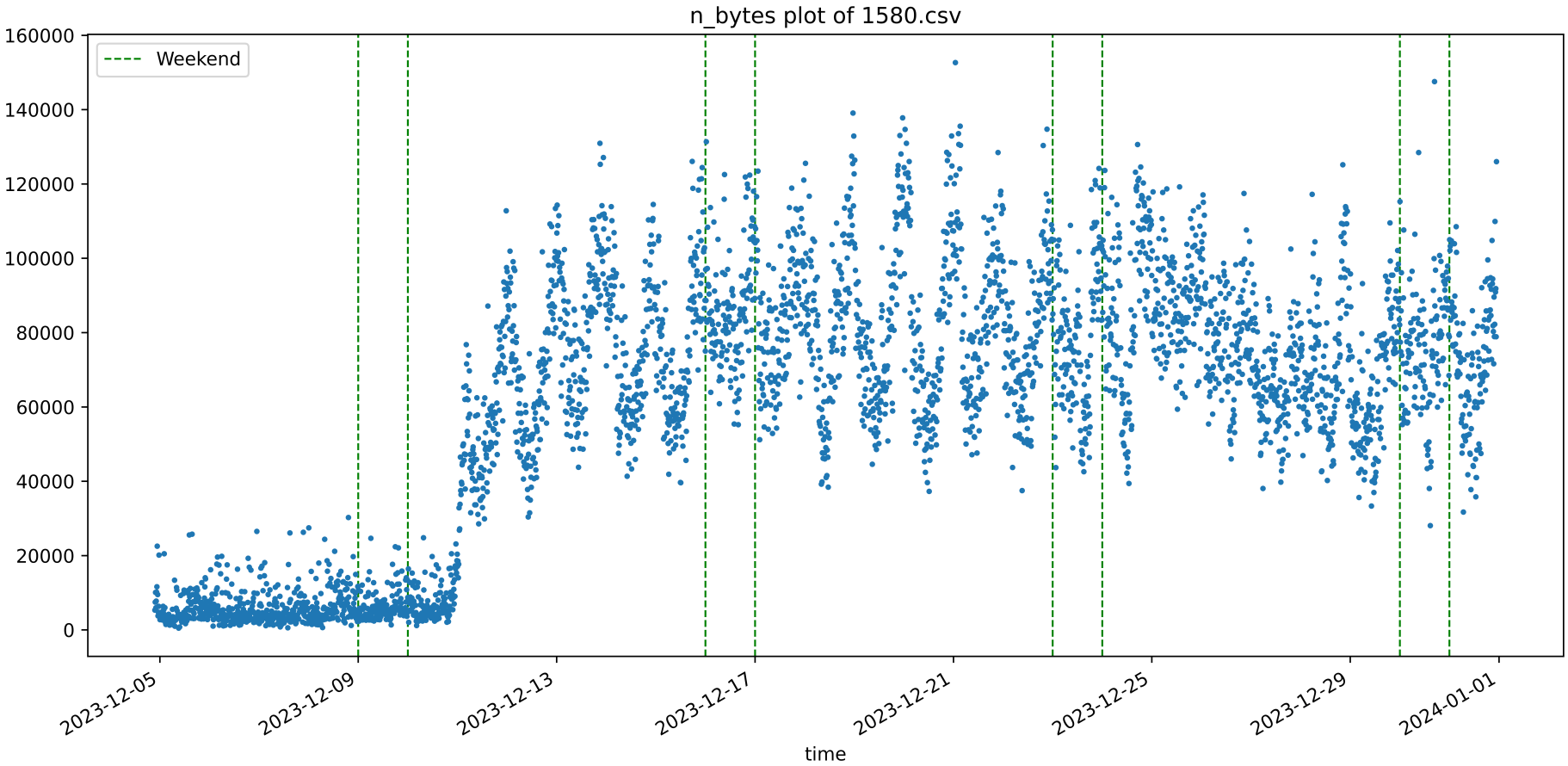




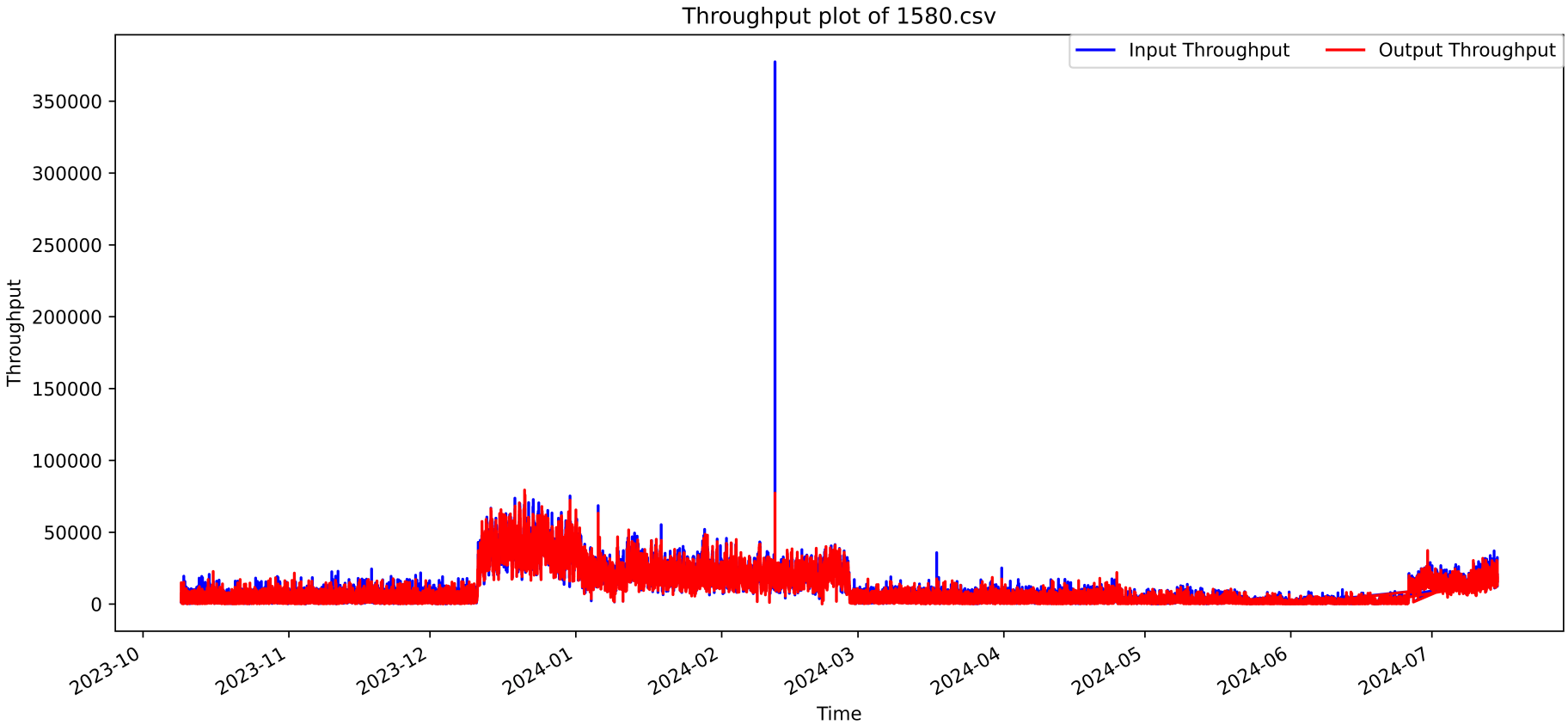




IP 1580 activity



IP 1580 throughput



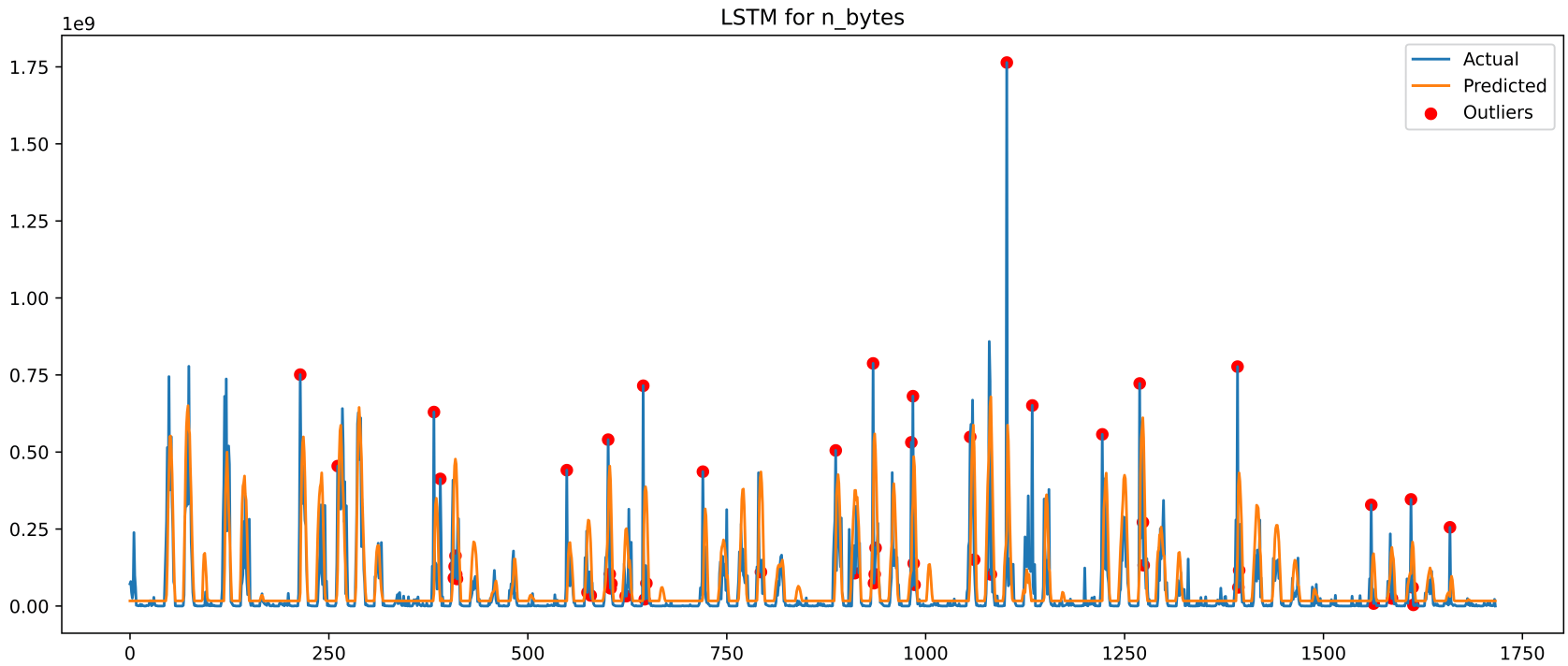
Anomalies detection methods

Model is based on provided notebook, except:

- Added another LSTM layer
- Added Dropout layer
- Smoother transition between LSTM and output layer

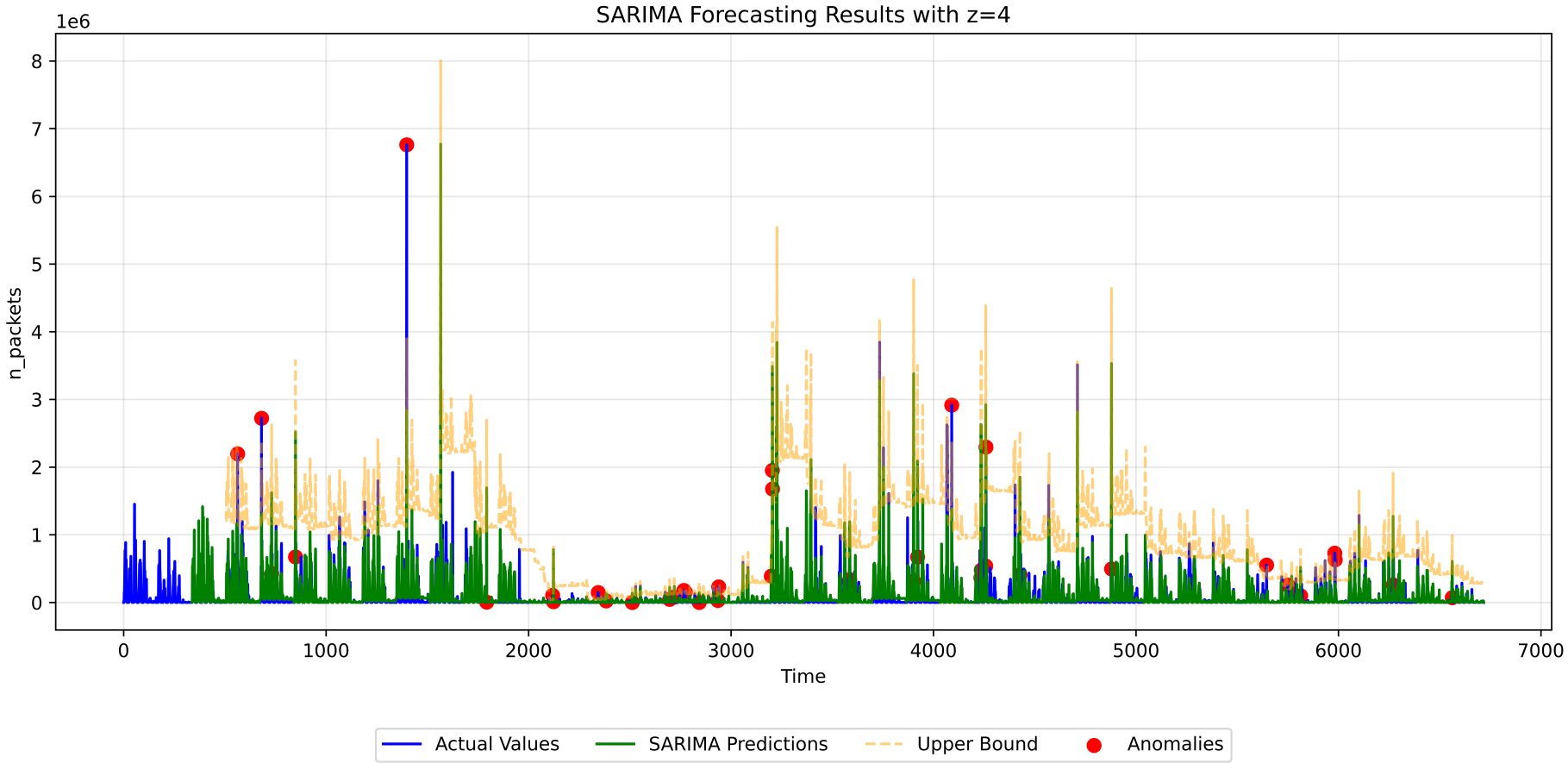


LSTM predictions on n_bytes

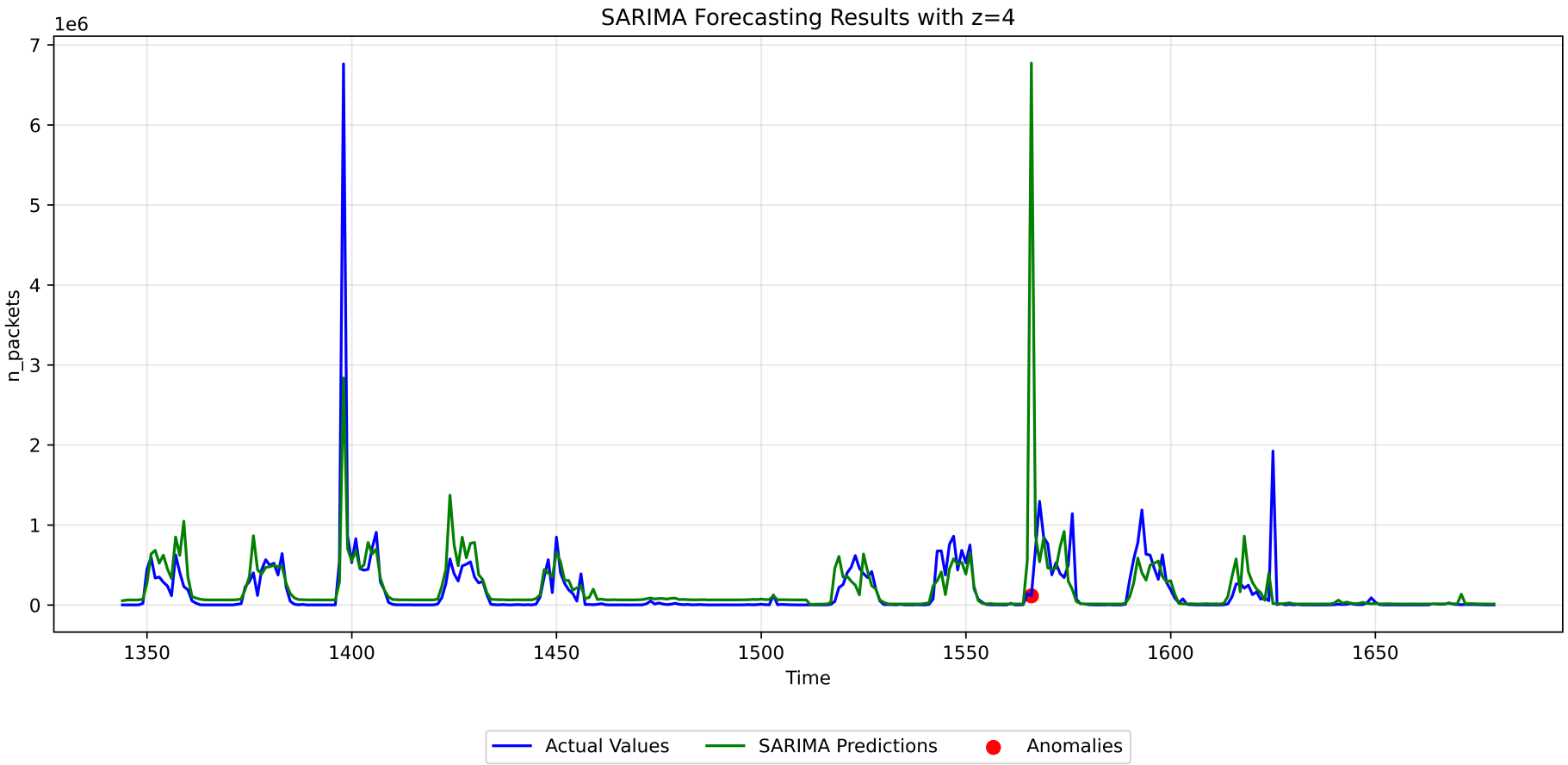


Metric	Training	Validation
Loss	0.3924	0.3028
R^2 Score	0.5118	-14.3551
RMSE	0.6157	0.5502
SMAPE (%)	50.8646	65.6712

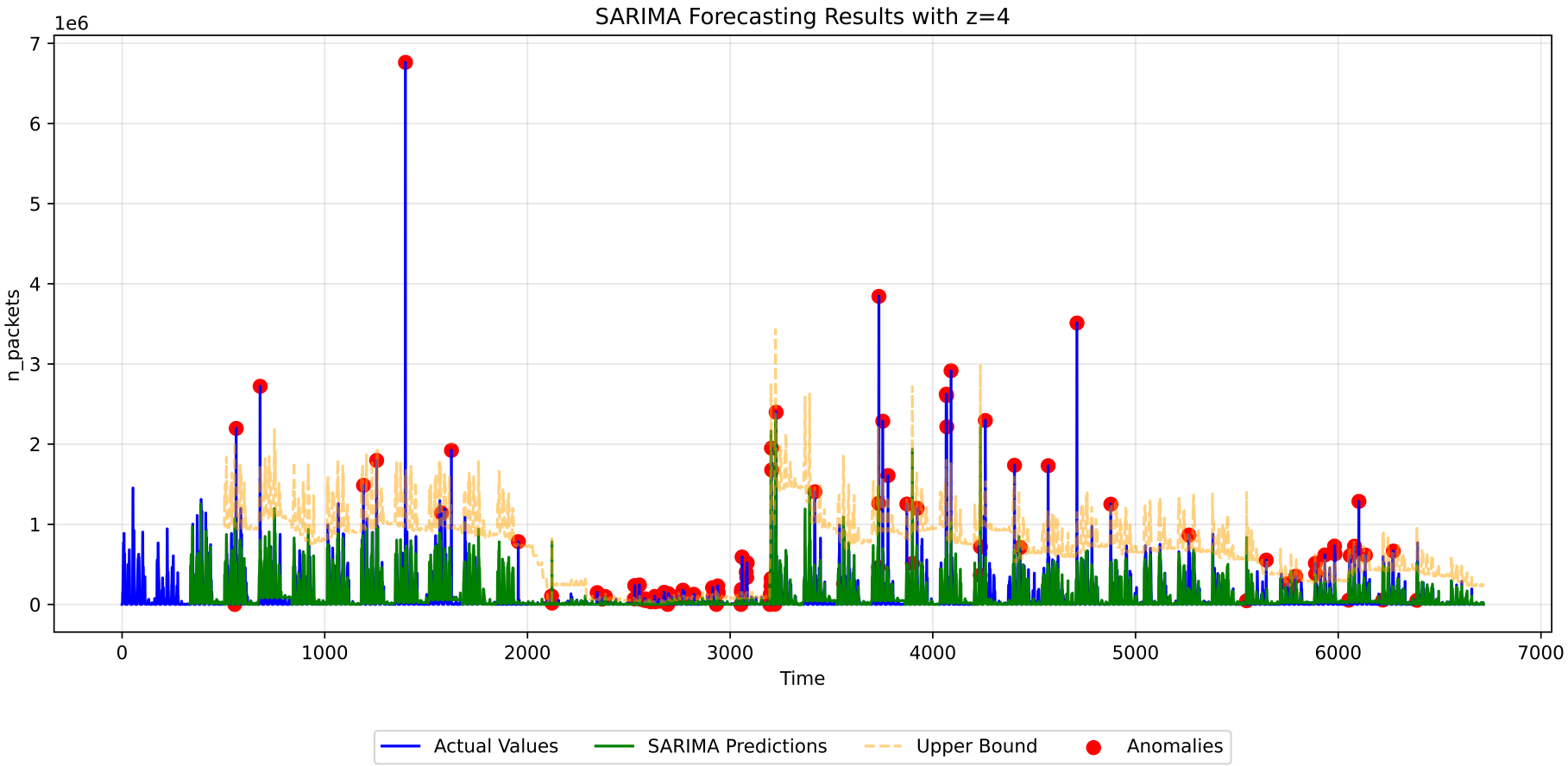
- Hourly aggregated dataset
- 168 hours period
- Overlapping 2 weeks window (by one week) problems: **outliers**



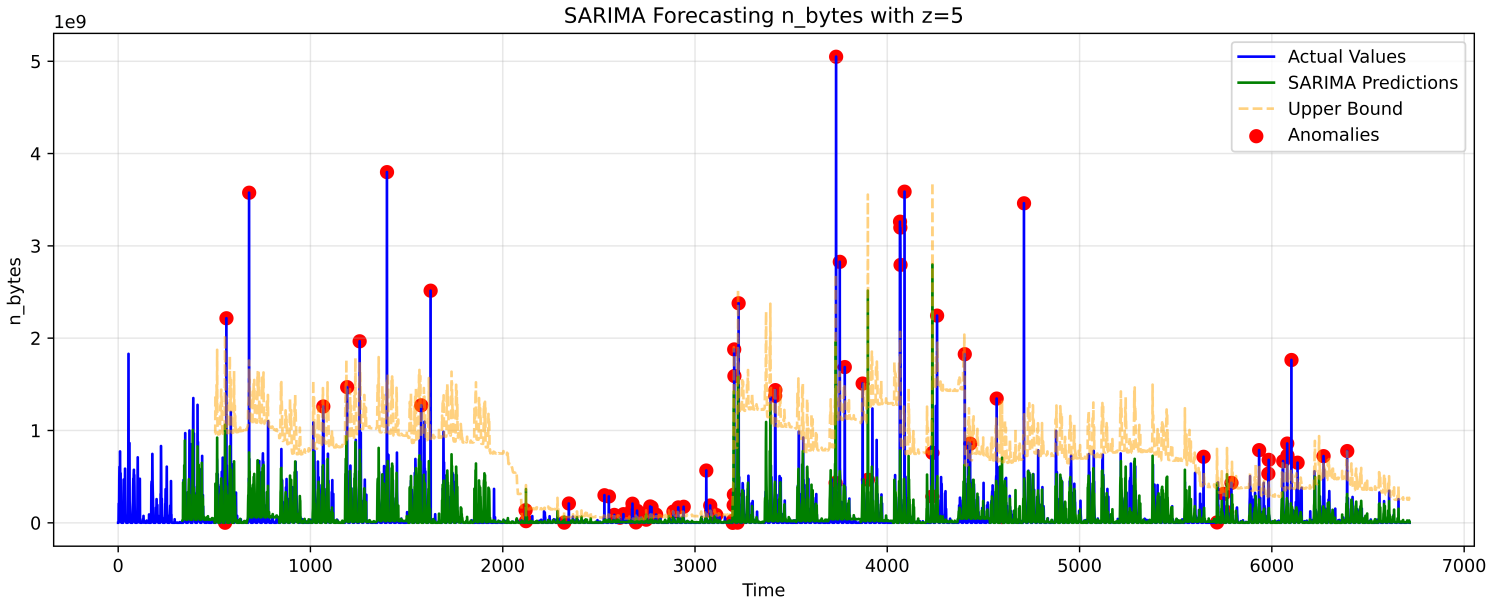
SARIMA - lagged outliers

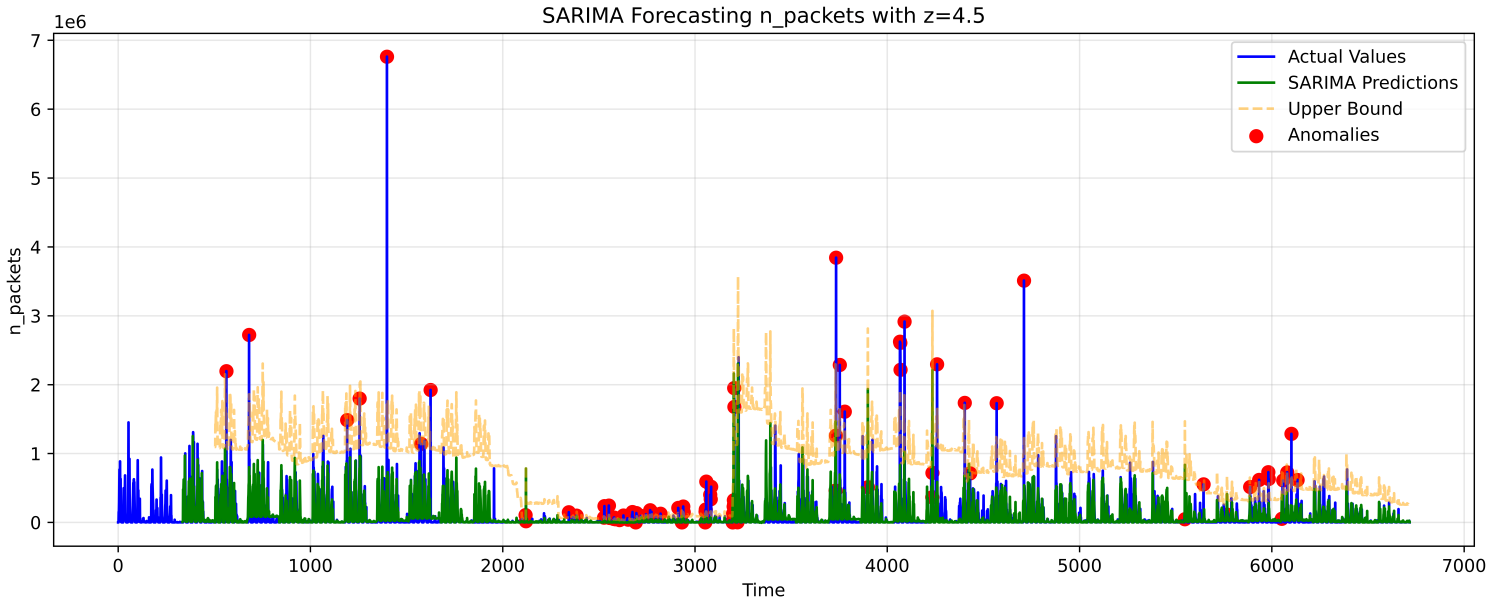


Solution: omitting outliers from previous training iteration



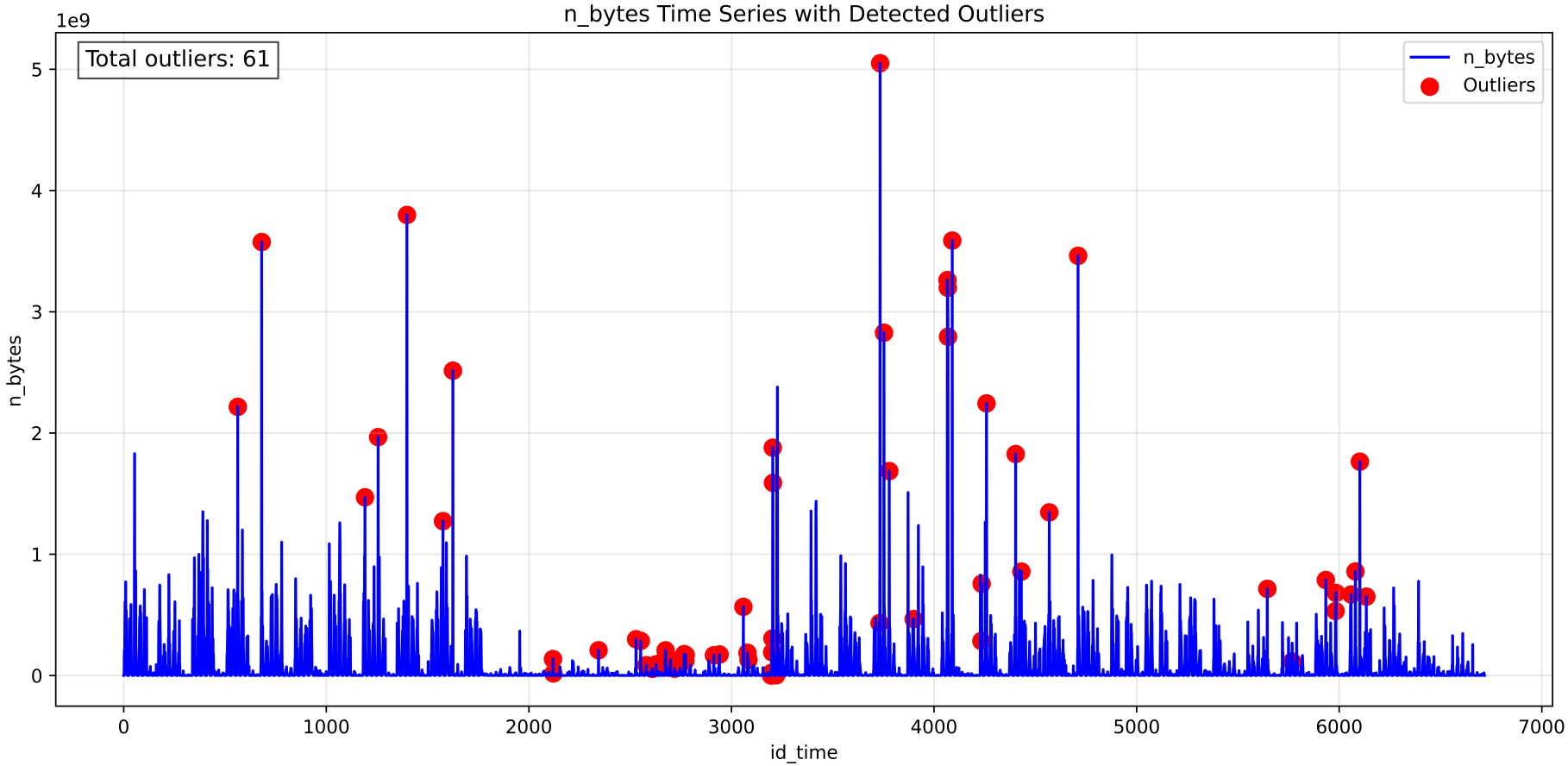
SARIMA - false positives





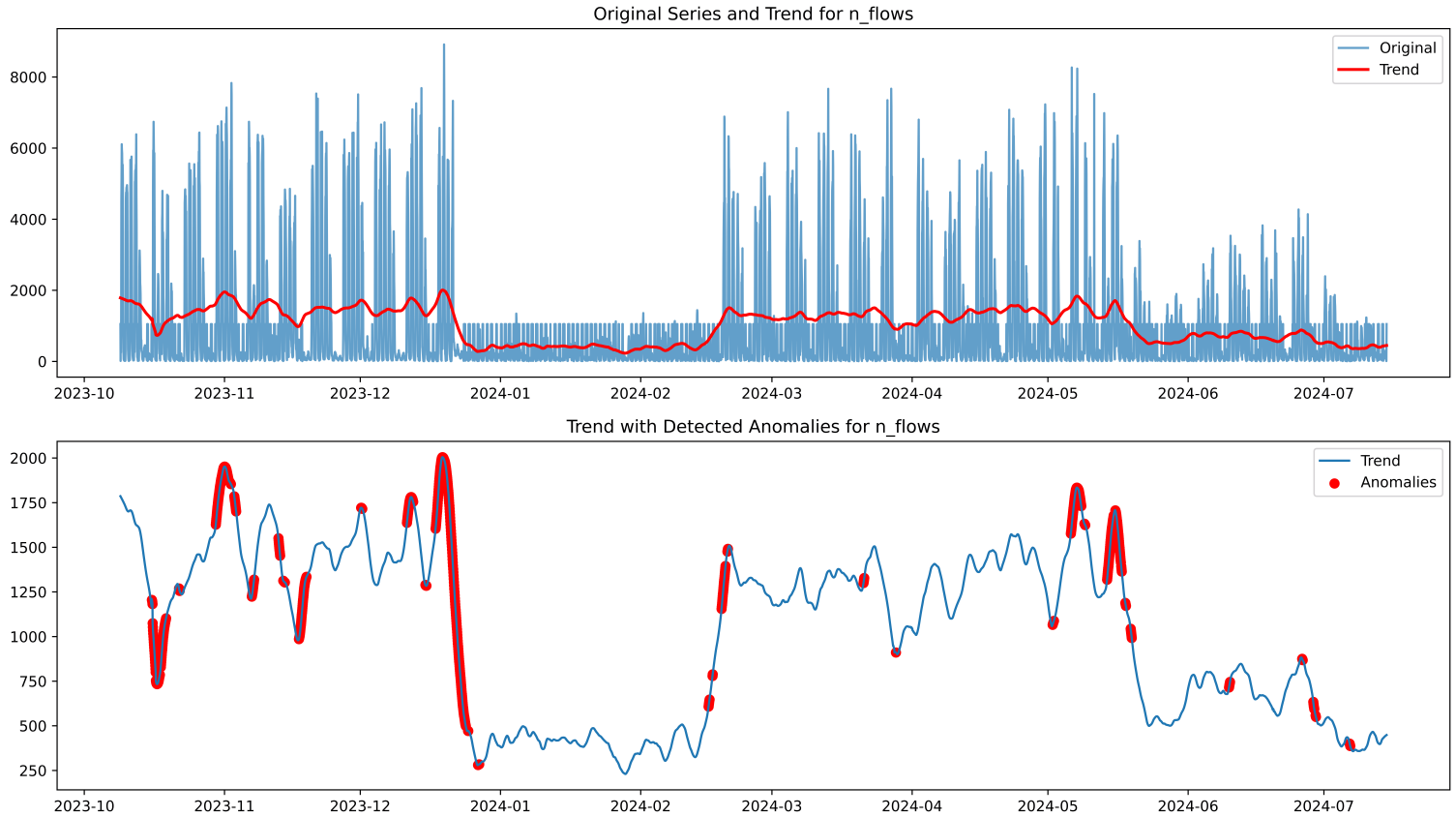
SARIMA - reducing false positives

Intersection of detected outliers across multiple metrics.
Reduced total amount of outliers from 111 to 60.



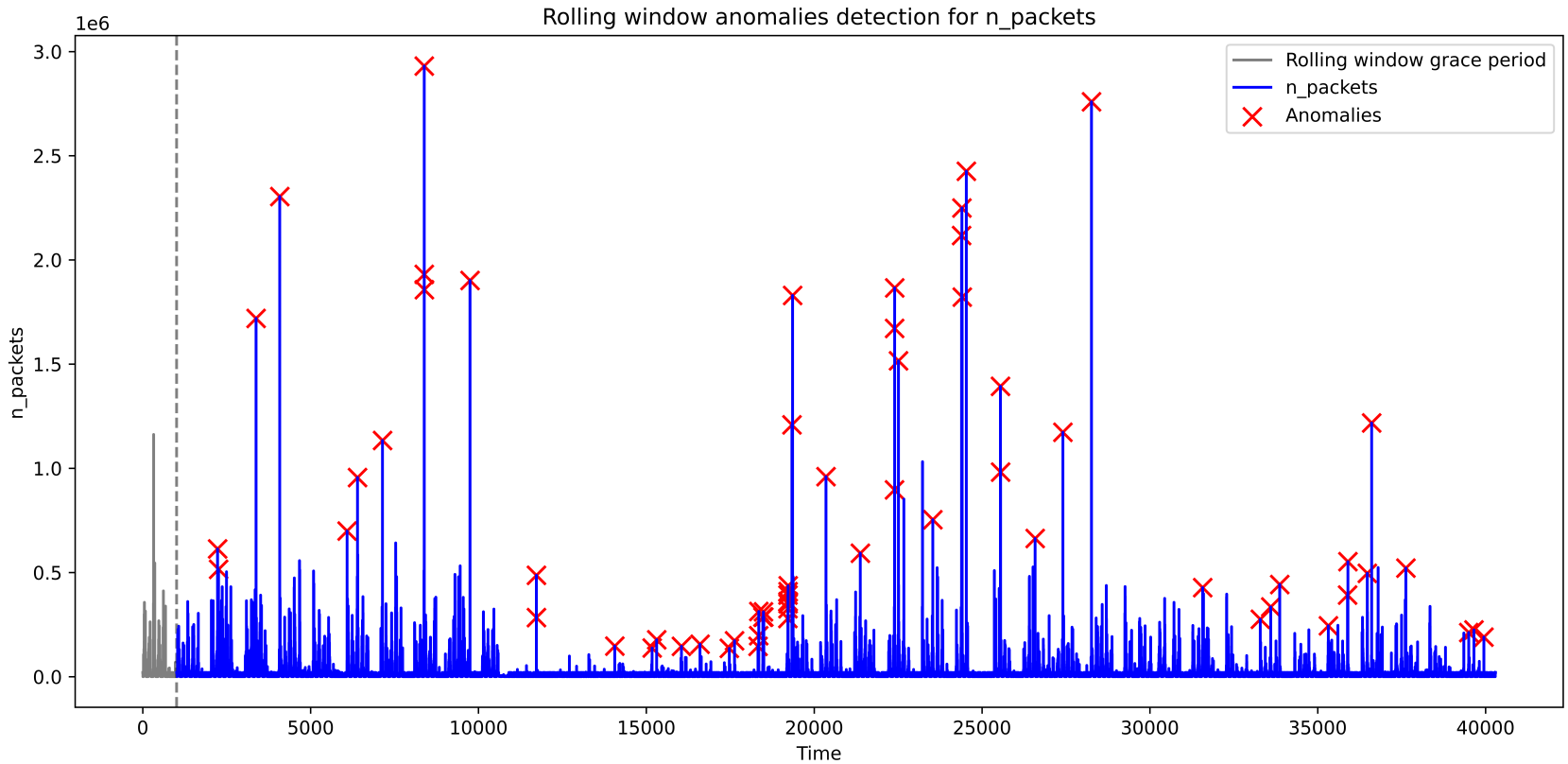
MSTL with Isolation Forest

- MSTL for trend, seasonality, and residuals decomposition
- Isolation forest trained on 8 hours rolling window
  - mean
  - difference
  - acceleration
- Adaptive to trends



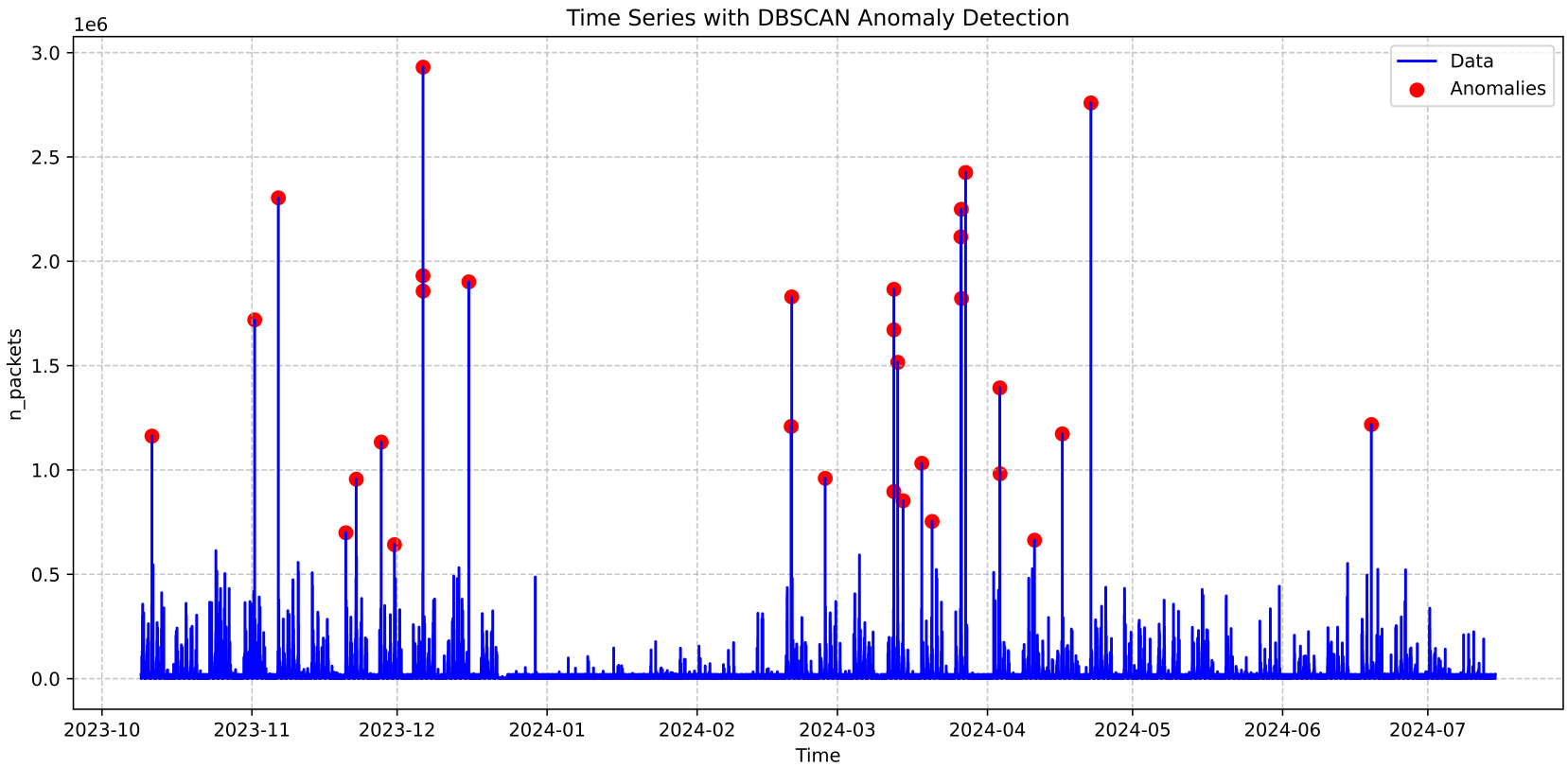
Rolling window with predefined threshold

- Simple, yet powerful.
- 1 week rolling window
  - mean
  - standard deviation
- Adaptive to trends
- Based on static threshold



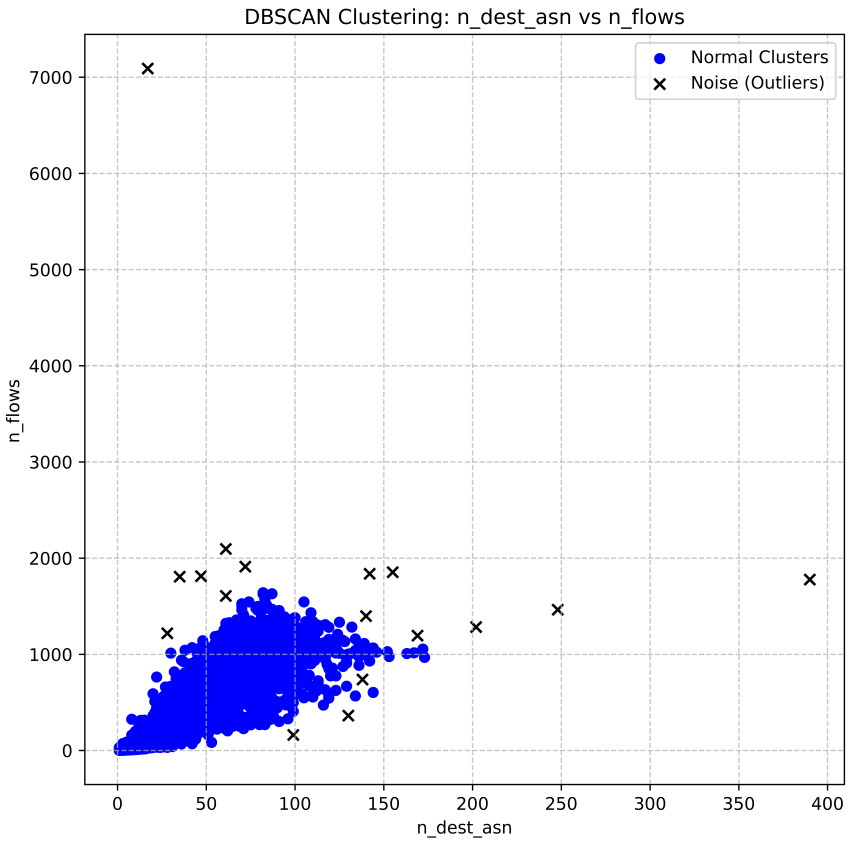
1D clustering - DBScan

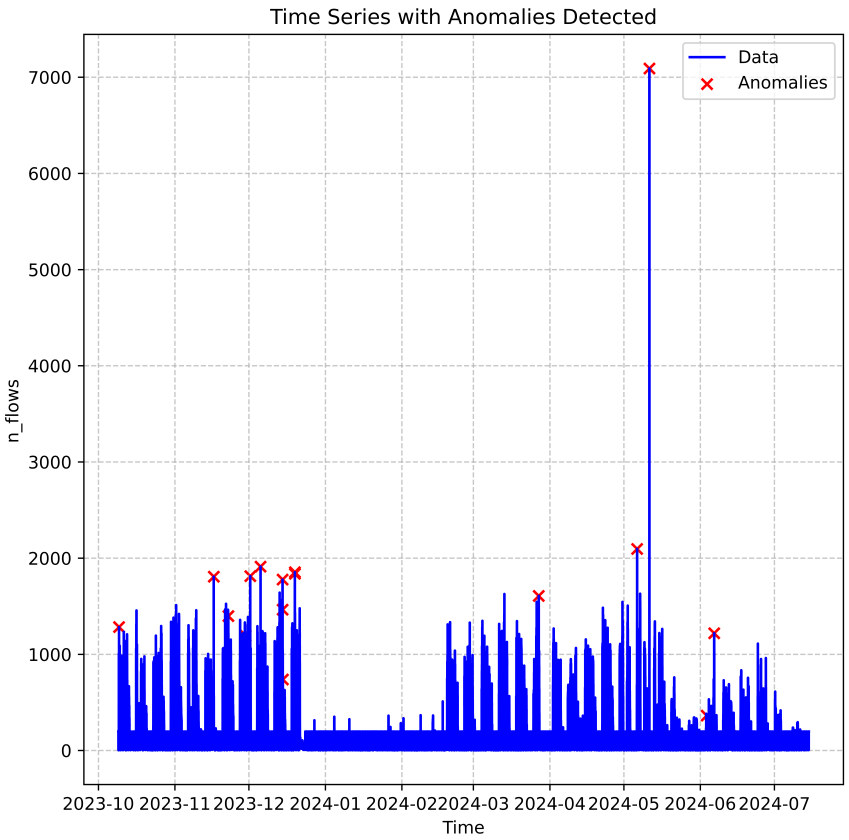
- Neighborhood based on metric
- Outliers based on Euclidean distance
  - outlying clusters
  - alone outliers
- Adaptive to trends



2D clustering - DBScan

- Neighborhood based on **2 metrics**
- Outliers based on Euclidean distance
  - outlying clusters
  - alone outliers
- Adaptive to trends





Thank You For Your Attention!